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Q. How does an operational amplifier work? Where is it used? And what are its advantages and disadvantages over other amplifiers?

Murali
Srikakulam

A. Operational amplifiers or op-amps are high-gain DC amplifiers with a negative feedback. Originally, these were used in circuits that performed various mathematical operations in analogue computers. Use of op-amps today, however, is not limited to mathematical operations, but

stage is an emitter follower, and the fourth stage is a DC-level translator and output driver. The arrangement is shown in Fig. 1.

Fig. 2 shows the standard symbol for an op-amp. There are two input terminals, one output terminal and a ground terminal which is usually not shown explicitly. The signal applied between the inverting input, marked with minus sign, and ground appears at the output with a 180° phase-shift. The signal applied between the non-inverting input, marked with plus sign, and ground appears at the output with a 0° phase-shift.

Many applications of op-amps require the use of only one input. Such an amplifier is called 'single-ended.'

The second (unused) input is either connected to other circuit elements or grounded. Other applications require that signals be applied to both the inputs so that the amplified difference of the two signals appears at the output. This configuration is called 'differential amplifier' or 'double-ended amplifier.'

Real vs ideal op-amp. Readily available, inexpensive op-amp ICs have characteristics which are reasonable approximations of an ideal op-amp as shown in the table.

Op-amp characteristics show that an op-amp is capable of providing

extends to control and instrumentation systems as well.

An op-amp usually consists of four stages in a cascaded form as shown in Fig. 1. The first stage is a differential amplifier with dual input (inverting and non-inverting) and double-ended output, which serves as input to the second stage of differential amplifier with a single-ended output. The third

what we expect from an ideal amplifier, in that its gain is infinite, its input impedance is very high (therefore it will not load the source which is driving it), its output impedance is low (therefore it will not get loaded when it is acting as a source) and, lastly, its bandwidth is very large. Therefore an op-amp is a much better device than a transistor or field-effect transistor.

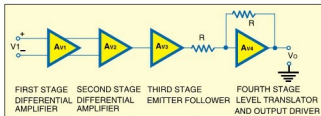


Fig. 1: Block diagram of an operational amplifier

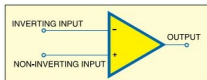


Fig. 2: Standard symbol of operational amplifier

Op-amp Characteristics

	Ideal op-amp	Typical op-amp
Gain (open-loop)	Infinite	10 ⁶ -10 ⁸
Input impedance	Infinite	2 mega-ohms
Output impedance	Zero	100-1000 ohms
Bandwidth	Infinite	1-5 MHz
Differential voltage gain	Infinite	10 ⁶ -10 ⁸
Common-mode voltage gain	Zero	20
CMRR	Infinite	100 dB
Slew rate	Infinite	1V/μs